

## Lecture 3: Lines and Planes in Space

### Background & Motivation

With vectors in hand, we can now describe geometric objects in 3D. A line needs a point and a direction; a plane needs a point and a normal. These are the simplest objects in space, and the techniques here—parametric equations, dot products for angles, cross products for normals—come back throughout the course.

### 1. Lines in $\mathbb{R}^3$

#### Definition 1: Parametric Equation of a Line

A line through  $P_0 = (x_0, y_0, z_0)$  with direction vector  $\mathbf{d} = \langle a, b, c \rangle$ :

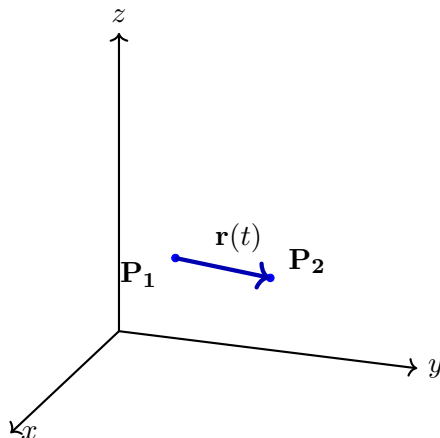
**Vector form:**

$$\mathbf{r}(t) = \underline{\hspace{2cm}}$$

**Component form:**

$$x = \underline{\hspace{2cm}}, \quad y = \underline{\hspace{2cm}}, \quad z = \underline{\hspace{2cm}}$$

– Each value of  $t \in (-\infty, \infty)$  gives a point on the line



The line segment connecting points  $\mathbf{P}_1$  and  $\mathbf{P}_2$  in 3D, represented by  $\mathbf{r}(t) = (1 - t)\mathbf{P}_1 + t\mathbf{P}_2$  for  $t \in [0, 1]$ .

### Definition 2: Line Segment

The segment from  $P_1$  to  $P_2$ :

$$\mathbf{r}(t) = \underline{\hspace{2cm}}$$

–  $t = 0$  gives  $P_1$ ;  $t = 1$  gives  $P_2$

### Example 1: Parametric Line

Write the parametric equation of the line through  $P_0 = (1, -2, 3)$  in direction  $\mathbf{d} = \langle 4, -1, 2 \rangle$ .

## 2. Planes in $\mathbb{R}^3$

### Definition 3: Equation of a Plane

A plane through  $P_0 = (x_0, y_0, z_0)$  with normal vector  $\mathbf{n} = \langle A, B, C \rangle$ :

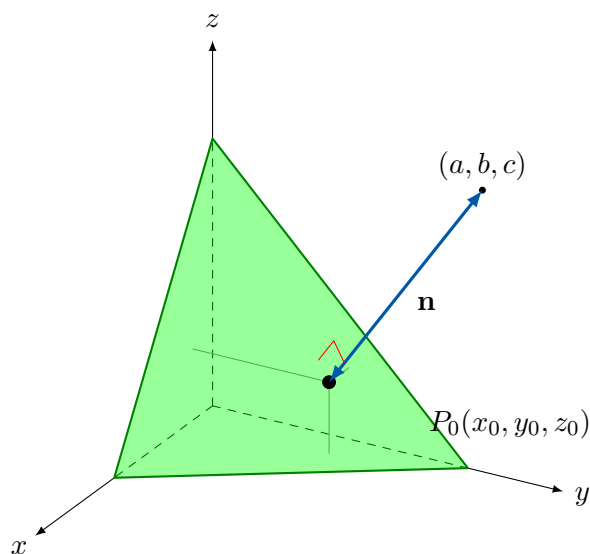
**Point-normal form:**

$$\underline{\hspace{2cm}}$$

**General form:**

$$\underline{\hspace{2cm}}$$

where  $D = Ax_0 + By_0 + Cz_0$



A plane in 3D space defined by a point  $P_0$  and a normal vector  $\mathbf{n}$  extending from  $P_0$  in the perpendicular direction.

### Example 2: Plane from Point and Normal

Find the equation of the plane through  $(2, -1, 3)$  with normal  $\mathbf{n} = \langle 4, 5, -2 \rangle$ .

### Example 3: Plane from Three Points

Find the equation of the plane through  $P_1 = (1, 2, 3)$ ,  $P_2 = (4, -1, 2)$ ,  $P_3 = (-2, 3, 5)$ .

## 3. Intersections and Angles

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### Line–Plane Intersection

– Strategy: substitute parametric equations of the line into the plane equation, solve for  $t$

### Example 4: Line–Plane Intersection

Find where  $\mathbf{r}(t) = \langle 1 + 2t, 2 - t, 3 + t \rangle$  intersects  $3x - y + 2z = 5$ .

### Angle Between Two Planes

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If planes have normals  $\mathbf{n}_1$  and  $\mathbf{n}_2$ :

$$\cos \theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|}$$

– Parallel:  $\mathbf{n}_1 \parallel \mathbf{n}_2$     – Perpendicular:  $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$

### Line of Intersection of Two Planes

#### Strategy

Given planes  $\Pi_1$  and  $\Pi_2$  with normals  $\mathbf{n}_1$  and  $\mathbf{n}_2$ :

1. **Direction:**  $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$
2. **Point:** Set one variable (e.g.  $z = 0$ ) and solve the  $2 \times 2$  system

#### Example 5: Intersection of Two Planes

Find the line of intersection of  $2x - y + z = 3$  and  $x + y - 2z = -4$ .

## 4. Distances

#### Theorem 1: Distance from a Point to a Plane

Distance from  $P_1 = (x_1, y_1, z_1)$  to  $Ax + By + Cz = D$ :

$$d = \frac{|Ax_1 + By_1 + Cz_1 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

#### Example 6: Point-to-Plane Distance

Find the distance from  $Q = (1, -1, -3)$  to the plane  $x + 2y + 3z = 18$ .

## 5. Vector Projection onto a Plane

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### Projection onto a Plane

To project  $\mathbf{a}$  onto a plane with normal  $\mathbf{n}$ :

$$\mathbf{a}_{\text{plane}} = \underline{\hspace{2cm}}$$

– Subtract the normal component from the vector

### Example 7: Projection onto a Plane

Project  $\mathbf{a} = \langle 3, 4, 5 \rangle$  onto the plane with normal  $\mathbf{n} = \langle 1, 1, 1 \rangle$ .

*Show all work for full credit.*

### Practice Problems

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**Problem 1.** Write the parametric equation of the line through  $P = (2, 1, -3)$  in direction  $\mathbf{d} = \langle 1, -2, 4 \rangle$ .

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**Problem 2.** Find the equation of the plane through  $(3, 0, -1)$  with normal  $\mathbf{n} = \langle 2, 1, -5 \rangle$ .

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**Problem 3.** Find where the line  $\mathbf{r}(t) = \langle -t, 2t, t + 3 \rangle$  intersects the plane  $-x + y = 6$ .

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**Problem 4.** Find the distance from  $(2, 3, -1)$  to the plane  $x - 2y + 2z = 4$ .

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**Problem 5.** Given  $\Pi_1 : x + 2y - z = 3$  and  $\Pi_2 : 2x - y + 3z = 1$ : (a) find the angle between the planes, (b) find the line of intersection.

Answer (a):  $\theta =$  \_\_\_\_\_

Answer (b):  $\mathbf{r}(t) =$  \_\_\_\_\_